

Quiz 5 (Set A)

Full Marks: 15

Duration: 20 minutes

Name:	ID:	Section:
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1. From the following scenarios, find the best suited transmission medium: [6 marks]
- You are switching the television channels with a remote.
 - FM 96.4 radio station is streaming your favorite song.
 - The dish cable in your house.
 - A network service company is providing about 1500 km long-haul data transmission.
 - A CAT 2 cable for telephone communication.
 - A cable for PC to PC connection.

- Infrared
- Radiowave
- Co-axial Cable
- Optical Fibre
- UTP
- UTP - Copper cross-over

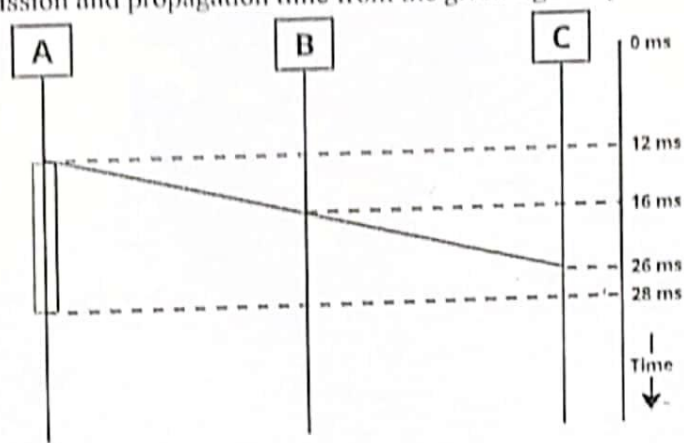
2. State the differences between pure ALOHA and slotted ALOHA. [3 marks]

3. Transmission time = $28 - 12 \text{ ms} = 16 \text{ ms}$
Propagation time = $26 - 12 \text{ ms} = 14 \text{ ms}$

4. $T_{fr} = \frac{500 \text{ bit}}{100 \text{ kbps}} = 5 \text{ ms}$
 $\therefore \text{vulnerable time} = 10 \text{ ms}$

5. Minimum $T_{fr} = 2 \times T_p = 2 \times 12.8 \mu\text{s} = 25.6 \mu\text{s}$
minimum size of frame = $5 \text{ Mbps} \times 25.6 \mu\text{s} = 128 \text{ bits}$
 $= 16 \text{ bytes}$

3. Find out the transmission and propagation time from the given figure. [2 marks]



4. A pure ALOHA network transmits 500-bit frames on a shared channel of 100 kbps. What is the requirement to make this frame collision-free? [2 marks]

5. A network using CSMA/CD has a bandwidth of 5 Mbps. If the maximum propagation time is $12.8 \mu s$, what is the minimum size of the frame in bytes? [2 marks]

Quiz 5 (Set B)

Full Marks: 15

Duration: 20 minutes

Name:	ID:	Section:
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1. From the following scenarios, find the best suited transmission medium: [6 marks]

- A cable for router to switch connection.
- A CAT 3 cable used in telecommunication.
- A 500 km long subscriber loop.
- Cable TV connection in your house.
- Bangladesh Betar is broadcasting your favourite show.
- You have bought a new wireless keyboard.

- UTP - copper straight-through
- UTP
- Optical fibre
- Co-axial cable
- Radio-wave
- Infrared

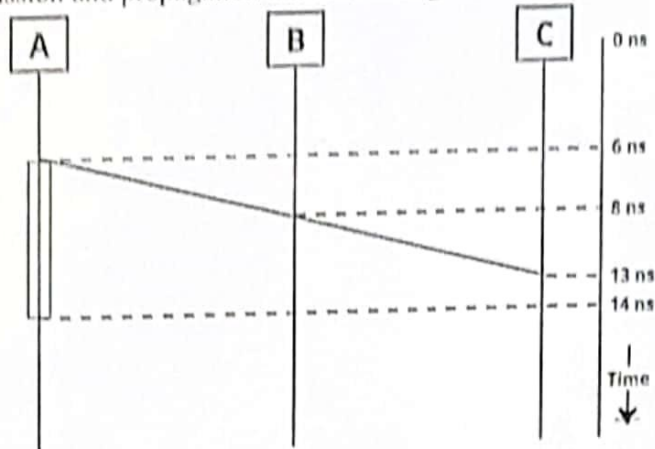
2. State the differences between pure ALOHA and slotted ALOHA. [3 marks]

3. Transmission time = $14 - 6 = 8 \text{ ns}$
Propagation time = $13 - 6 = 7 \text{ ns}$

4. $T_{fr} = \frac{800 \text{ bit}}{2 \text{ Mbps}} = 400 \text{ ns} \rightarrow 4 \times 10^{-4} \text{ s}$
vulnerable time = $800 \text{ ns} \rightarrow 8 \times 10^{-4} \text{ s}$

5. Minimum $T_{fr} = 2 \times T_p = 2 \times 51.2 \text{ ns} = 102.4 \text{ ns}$
minimum size of frame = $\frac{20 \text{ Mbps}}{102.4 \text{ ns}}$
 $20 \text{ Mbps} \times 102.4 \text{ ns} = 2048 \text{ bits}$
 $= 256 \text{ bytes}$

3. Find out the transmission and propagation time from the given figure. [2 marks]



4. A pure ALOHA network transmits 800-bit frames on a shared channel of 2 Mbps. What is the requirement to make this frame collision-free? [2 marks]

5. A network using CSMA/CD has a bandwidth of 20 Mbps. If the maximum propagation time is 51.2 μ s, what is the minimum size of the frame in bytes? [2 marks]